

Smart Scan Strategy for Electronic Warfare

Technical explainer — the hardware constraint, the model lineup, and the test suite

Domain EW receiver scan scheduling Policies 7, three tiers Scenarios 7 synthetic + real recorded radar data

01

The hardware constraint — the actual problem

An EW receiver cannot listen to the whole spectrum at once. Every decision this project makes exists because of three physical limits on the receiver itself, not the difficulty of detection.

K = 2

bands observable at once, out of 16–32 possible bands depending on scenario — 6–12% of the spectrum visible at any instant

3 ms

tuning delay: dead time paid every time the receiver retunes to a new band, out of a 10 ms decision step

4 ms

minimum dwell: stay tuned less than this and the step yields zero usable signal — scanning too fast means seeing nothing

These three numbers do not change across any scenario in this project. The receiver hardware is held constant on purpose. What changes is the threat environment around it. So the entire problem the models are solving is never "detect a signal" — the detection physics are simple and fixed. It is: **out of a spectrum you mostly cannot see, where do you point this narrow window next, and for how long?** That is a scheduling problem, not a sensing problem, and it is why the solution is a family of scheduling policies rather than a better detector.

02

The models — an escalation ladder, not a menu

Seven policies across three tiers. Each step up in complexity has to earn its place with evidence, because a heavier model costs more compute — a real constraint on hardware meant to eventually run in real time on the receiver itself.

Tier 1 — Reference

WHAT TODAY'S RECEIVERS DO

Baseline — a fixed round-robin sweep. Never looks at what it just found. This is the incumbent, non-adaptive approach the whole project exists to beat.

Tier 2 — Engineered, not trained

DOMAIN KNOWLEDGE, ZERO LEARNING

CTMC floor — a randomized walk, grounded in a real theorem (Clarkson & Pollington): against emitters with unknown behavior, no fixed deterministic strategy beats random. This is the safety net every other policy sits above.

Index — a hand-tuned scheduler weighing deadline pressure, novelty, and estimated per-band value. Real scheduling theory, no machine learning at all.

Tier 3 — Learned

THE ACTUAL AI/ML, CHEAPEST TO HEAVIEST

Bandit — the simplest learner: a linear model predicting how much scanning a given band is worth right now, updated online. Trains in minutes on a CPU.

Q-learning — adds memory of consequences. Can learn "this band is quiet now but matters soon," which the bandit structurally cannot represent since it only ever predicts immediate payoff.

DQN / PPO — full neural networks. Deliberately the last tier built, gated behind the simpler models already proving the approach works — reaching for deep RL by default is treated as a risk to avoid, not a default.

Why keep all seven instead of shipping one winner: the finding itself is the point. On real recorded radar data, the cheapest learner (bandit) landed within a few points of the most expensive one (PPO). For hardware that has to run in real time on constrained compute, that is the headline, not a footnote — the expensive model is not required to capture most of the value.

03

The scenarios — same hardware, harder enemies

Every scenario below keeps the identical receiver constraint (K = 2, same dwell, same retune cost). Only the threat environment changes — the correct way to stress-test a scheduling policy rather than a detector.

SCENARIO	SPECTRUM	WHAT IT STRESSES
A · Mostly Fixed	16 bands, 10 emitters	The easy case — can a learned policy still avoid wasting time even when nothing is tricky?
B · Mostly Periodic	16 bands, 10 emitters	The headline scenario. Emitters blink on and off in a rhythm; blind round-robin ends up polling during the OFF phase over and over. Rewards a policy that learns the rhythm.
C · Frequency-Agile	24 bands, 12 emitters	Emitters hop frequency to evade — tests whether the policy can track a moving target.
D · Mixed Environment	24 bands, 15 emitters	An even mix of every behavior type — the generalization check.
E · High-Density	32 bands, 30 emitters	So much is active at once that almost any scan finds something — tests precision (avoiding false alarms), not just raw detection.
F · Sparse	32 bands, 5 emitters	A needle in a haystack — the largest expected margin for a smart policy over random.
G · Rapidly Changing	24 bands, 15 emitters	Emitters switch behavior mid-run — tests whether the policy adapts or clings to a stale read of the environment.

04

Validation on real data

Beyond the seven synthetic scenarios, every policy was re-evaluated against real recorded radar pulse data — the Turing Synthetic Radar Dataset, the same dataset the DRDO problem statement itself points to — not another synthetic scenario, but actual physical pulse recordings, filtered down to realistic scanning-mode files and evaluated with priority weighting derived from real received signal amplitude.

The synthetic-scenario story mostly held up. One genuine surprise did not: DQN and PPO, which underperformed the simpler learners badly on synthetic data, turned out to be the **strongest** performers once run against real signal statistics — PPO ended up the best policy overall on real data, of all seven. That reversal was tested for and confirmed, not assumed.

Generated as reference material for a technical walkthrough of the SIH26055 solution. Figures reflect direct evaluation runs against the project's own scenario configs and the Turing Synthetic Radar Dataset (Apache-2.0, Alan Turing Institute).